

Bioremediation of Wastewater

Effect of Algae in bioremediation of Nitrate and Phosphate content in Wastewater.

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Introduction:

The world is facing a huge problem related to the demand and quality of water. On one hand, there is little availability of drinking water and, on the other, management and remediation of wastewater becomes a huge challenge for the developed and developing countries. The growing population of the world and the progressive adoption of an urban, industrial lifestyle have deteriorated the quality of freshwater reservoirs around the world and has caused freshwater shortages in many areas. Wastewater is any form of water that has been contaminated by commercial or domestic processes. This includes water from sewage and polluted water that is industrial byproduct. The characteristics of wastewater vary depending on the source. Types of wastewater include domestic wastewater from households/communities, sewage etc, Industrial wastewater from industrial activities, and agricultural wastewater. Wastewater can contain physical, chemical and biological pollutants including many harmful substances and cannot be released back into the environment until it is treated. Thus, the importance of wastewater treatment is twofold – a) To restore the water supply and b) To protect the environment from toxins.

Environmental remediation deals with the removal of pollution or contaminants from environmental media such as soil, groundwater, sediment, or surface water for the general protection of human health and environment. Traditional Wastewater treatment plants are effective in remediation of suspended solids and many harmful elements, but the Toxic heavy metals, Nitrate and Phosphate content in wastewater removal processes still are not very effective and sustainable, causing those toxic elements being discharged to ground water. Bioremediation can be effectively used for these. It uses naturally occurring microorganisms (yeast, fungi, algae, bacteria) to break down, or degrade, hazardous substances into less toxic or nontoxic substances. Hence it is a natural way to remediate those toxic elements.

In the current project, it aims to determine the effectivity of algae to bioremediate the Nitrate and Phosphate content from wastewater. Algae absorbs nitrogen in form of nitrate and phosphorus in form of phosphate from the wastewater and clean it by separating the nutrients as biomass. This algal biomass can also be used later as an energy source, turned into biofuels/biodiesel, which is renewable and has a small carbon footprint.

Question, Hypothesis and Variables:

Question:

How do different types of algae affect the bioremediation of wastewater by reducing the nitrate and phosphate concentration in the wastewater over time?

Hypothesis:

My hypothesis is that if 4 different strains of algae

- i. *Chlamydomonas reinhardtii*
- ii. *Arthrospira platensis* (Spirulina)
- iii. *Scenedesmus quadricauda*
- iv. *Chlorella vulgaris*.

is used to bioremediate wastewater by removing the Nitrate and Phosphate concentration in wastewater, *Chlamydomonas reinhardtii* would be most effective in the bioremediation as it is believed to absorb the most Nitrate and Phosphate in its biomass and use that for photosynthesis and green algae production. However, all the 4 types of algae specimens will be able to remove the Nitrate and Phosphate concentration from wastewater as Nitrate and Phosphate are nutrients which favored among plants and is known to stimulate green algae (autotrophic organisms) production.

Independent variable:

1. 4 different streams of Algae.

Dependent variable:

1. Phosphate concentration in the wastewater.
2. Nitrate concentration in the wastewater.

Constants:

1. Volume of algae culture from starter culture
2. Volume of wastewater sample.
3. Volume of air
4. Temperature
5. Light

Background Research:

What is wastewater:

Wastewater is essentially the water supply that is used for many applications. It can be defined as a combination of liquid or water carried waste removed from residence, institution and commercial/industrial establishment together with such ground water, surface water and storm water as may be present. Thus, Wastewater is any type of water that has been negatively affected in quality as it is contaminated by commercial or domestic processes.

There are mainly 3 kinds of wastewater:

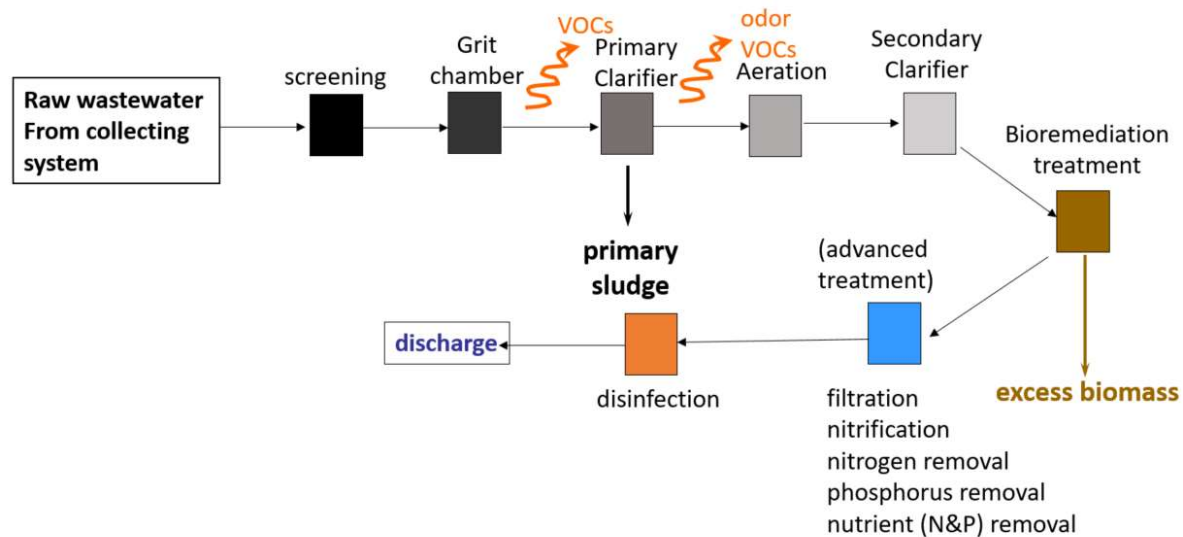
- a) Domestic wastewater.**
- b) Industrial wastewater.**
- c) Agricultural Wastewater.**

Domestic wastewater consists of domestic wastewaters originated from households, sewer overflows and storm drains. On average, each American uses about 150 gallons of water a day. Most of it in the home and produces 80 gallons of wastewater per day. The total domestic wastewater generation rate in the US is about 31×10^9 million gallons per day. Remediating at least half volume of the wastewater generated daily will provide a solution for water crisis.

Different industries produce various kinds of industrial wastewater based on their own specific combination of pollutants. Among those, metal working industry serves as an important pollution distributor discharging heavy metal waste (toxic waste) such as nickel, lead, chromium, cadmium, zinc and iron contaminated wastewater into the environment. This indiscriminate disposal of industrial and domestic wastes leads to heavy pollution of water, threatening all kinds of inhabiting organisms. Therefore, it is necessary to focus on the treatment techniques and ways to partially remove contaminants (nitrates, phosphates and heavy metals) from wastewater before discharging them into waterways.

Conventional Wastewater Treatment:

Most of the wastewater treatment plants employ conventional wastewater treatment methods to reduce the contaminant concentration and to improve the quality of wastewater effluent before it discharges to groundwater or re-enters water bodies. This method is employed only for treating municipal wastewater and is not designed and equipped for handling toxic industrial waste. Toxic waste is pretreated at the source of generation (industries) bringing down the concentration level and is sent to wastewater treatment plant as industrial wastewater. Figure 1 below shows the schematic representation of conventional wastewater treatment plant.



Schematic representation of wastewater treatment

Figure 1: Schematic representation of wastewater treatment plant

In brief, municipal and industrial wastewater is initially collected and subjected to two main treatment processes: primary and secondary treatment. During primary treatment, pretreatment is performed which involves the usage of bar screens to remove large solids from the wastewater. The water is then pumped to pre-aeration tanks where grit (sand, eggshells, etc.) is removed. Next, the wastewater is fed to a primary clarifier where suspended solids are allowed to settle

(primary settlement). After the primary clarifier, the wastewater still contains high amount of organic content and is sent to secondary treatment. During the secondary treatment, the organic content of the wastewater is degraded by microorganisms within the aeration tanks, and waste sludge is allowed to settle in the secondary clarifier. Sludge that settles on the bottom of the primary and secondary clarifiers is pumped to a digester via a sludge thickener to produce methane and carbon dioxide. However, the water leaving the secondary clarifier could still contains significant amounts of nitrogen and phosphorus. Also, in this process, there is no special method employed during the treatment cycle to remove heavy metals from wastewater.

Nitrogen and Phosphorus Contamination:

Conventional wastewater treatment method can only reduce nitrogen and phosphorus concentrations in wastewater to a limited extent. Discharge of this partially treated water leads to high release of nitrates and phosphates into the environment. In addition, agricultural activities also contribute significantly for the release of nitrates and phosphates to the groundwater in most parts of the United States. They contribute about 50% of the nitrogen and phosphorus entering surface waters. Livestock and dairy farm operations also contribute their share to nitrate and phosphate loading of soils and contamination of groundwater due to the mobility of nitrates and phosphates through the soil. Storm water run-off and chemical pollution (automotive products, paint, cleaners, and pesticides) are the other sources that release nitrogen and phosphorus into the environment.

Bioremediation and it's types:

To overcome these problems, there is a need for economic and energy efficient nitrogen, phosphorus and heavy metal removal technology. Bioremediation by using microorganisms for the treatment of metal enriched wastewaters containing high amounts of nitrates and phosphates can be effectively used to mitigate these issues and provides a means for cost-effective removal of contaminants in wastewater.

Depending of type of microorganisms used, bioremediation can be of different types.

- a) Native when microorganisms that are found already living at a given site are stimulated to grow. The proper soil temperature, oxygen, and nutrient content may need to be provided.

- b) Nonnative when microorganisms from other locations, or genetically engineered microorganisms, can be added to the contaminated site, if they are proven to be efficient in degradation of specific pollutant.

Depending of the location also, bioremediation can be:

- a) In situ: Bioremediation treats the contaminated soil or groundwater in the location in which it is found.
- b) Ex situ: bioremediation processes require excavation of contaminated soil or pumping of groundwater before they can be treated in specific location in strict controlled conditions.

Algal bioremediation uses algae to "scrub" polluted waters and to transform that algal biomass into biofuel. It is a double win: clean water and regenerable source of energy.

What is Algae:

Algae are phototrophic unicellular organisms found individually or in groups that consume carbon dioxide, nitrogen and phosphorus, and release oxygen. Algae have an affinity for polyvalent metals and are very effective in removing heavy metals, nitrates and phosphates present in wastewater. Therefore, use of algae for bioremediation of wastewater offers potential advantages over other techniques in use. In recent times, algae has become of great interest since it provides several advantages such as biofuel production, carbon dioxide mitigation, wastewater treatment, food production and so on.

Wastewater Treatment with Algae:

Algae have a great potential to solve energy and environmental challenges around the world. Wastewater treatment with algae is a more environmental sound approach to reduce nitrogen and phosphorus and to remove heavy metals from wastewater. Algae can absorb significant amount of nutrients because they need large amounts of nitrogen and phosphorus for proteins (45-60% algae dry weight) and metals as micronutrients for their growth. Figure 2 shows the process involved in high rate algal pond in which algae plays a dual role by assimilating nutrients from wastewater and supplying oxygen to bacteria. The bacteria take up the oxygen and degrade organic material in the wastewater, the same process which is used in activated sludge treatment.

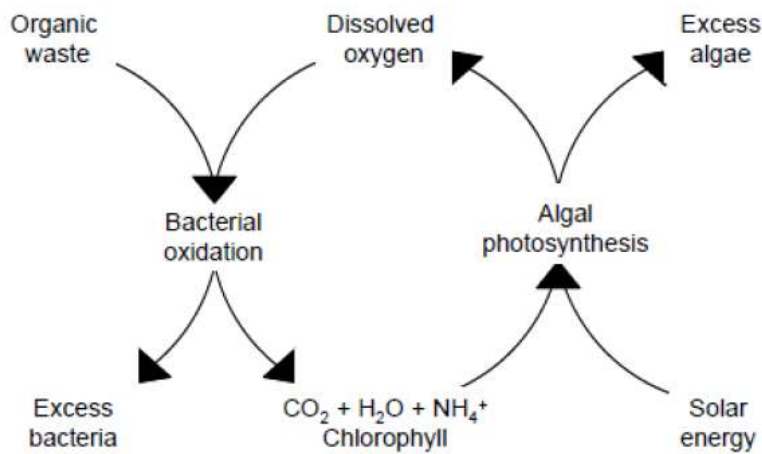


Figure 2: Algal bioremediation process.

The selection of algae strain to be used in wastewater treatment is determined by their robustness against wastewater and by their ability to grow in and to assimilate nutrients from wastewater. *Chlorella*, *Scenedesmus*, *Chlamydomonas* and *Spirulina* are the widely used algae species in experimental studies of wastewater treatment. Algae are aquatic microscopic plants or organisms with chlorophyll and body not differentiated into root, stem and leaf. Photosynthetic mechanism in algae is similar to land-based plants, except these are generally more efficient in converting solar energy into biomass. A much simpler cellular structure, an aqueous environment where they have efficient access to water, CO₂ and other nutrients, and a greater capacity to adapt makes algae ideal organisms to clean wastewater.

Nutrient removal by algae involves various metabolic pathways of the algal cell. Carbon, nitrogen, phosphorous and sulfur are the main components responsible for algal growth. Metals such as potassium, sodium, calcium, iron and magnesium serve as micronutrients for the growth of algae. Emphasis is usually put on nitrogen and phosphorus uptake but importance is also given to the technological aspects of heavy metal removal by algal biomass. Algae turns out to large biomass due to their high absorption capacity compared to other microbial bio-absorbents.

Chlorophyll and Photosynthesis:

Chlorophyll is green pigment with a complex structure containing magnesium, nitrogen, oxygen, hydrogen, carbon. There are different forms of chlorophyll, chlorophyll a is common to plants, which use photosynthesis, chlorophyll b is found in green algae and chlorophyll c is found in some primitive bacteria. A chlorophyll molecule has a hydrophobic "tail" that embeds the molecule into the thylakoid membrane. The 'head' of a chlorophyll molecule is a ring called a porphyrin. The porphyrin ring of chlorophyll, which has a magnesium atom at its center, is the part of a chlorophyll molecule that absorbs light energy.

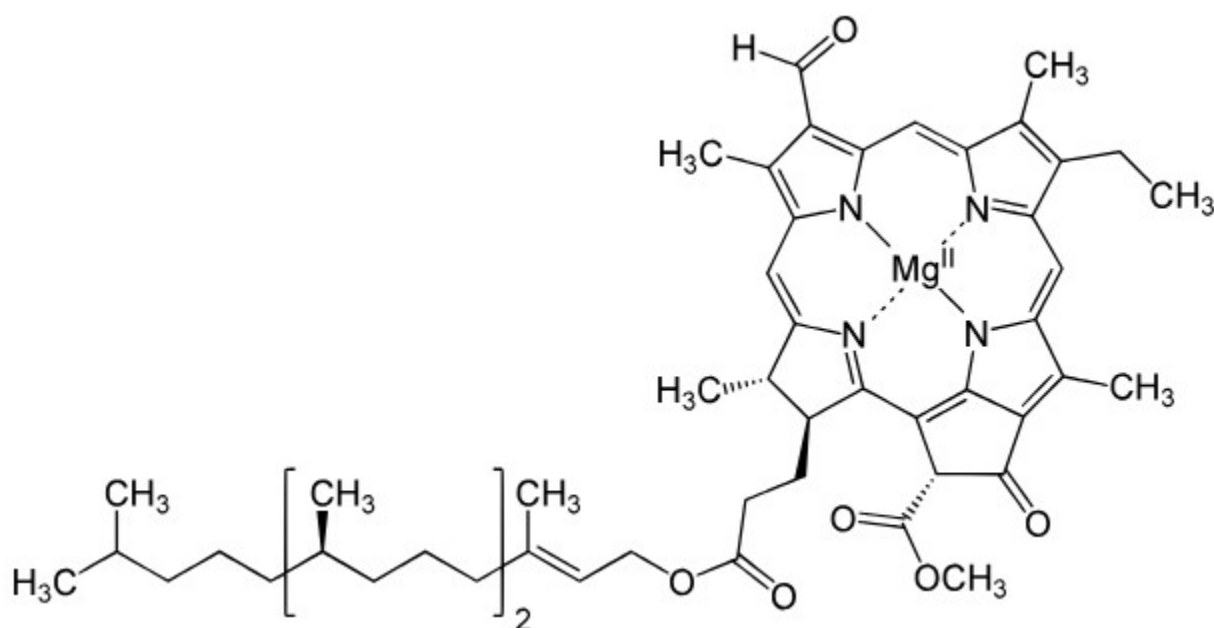


Figure 3: Chemical structure of Chlorophyll.

Chlorophyll absorbs sunlight energy and turns it into chemical energy in the process of photosynthesis. This energy transforms carbon dioxide and water in oxygen and carbohydrates. Chlorophyll absorbs all light form the spectrum expect green, which gives it its color.

When a chlorophyll electron absorbs light energy, it moves from a lower state of energy, to higher state of energy because of the light dependent processes, which converts the energy in ATP (Adenosine triphosphate). Oxygen is the by- product of this reaction. Carbon dioxide is captured from the water or atmosphere to form carbohydrates in the light reaction as well.

Photosynthesis is the process of water, sunlight and carbon dioxide being converted to carbohydrates, oxygen. This happens by converting energy of the sun into potential chemical energy. There are two stages for photosynthesis, the light reaction and the dark reactions. The light reaction requires a direct energy of light to make certain molecules for the dark reaction where the products of the first reaction are used to make bonds of carbohydrates.

Carbon fixation happens in the dark stage, where the energy from the previous stage (ATP) is used to combine hydrogen and carbon to make carbohydrates.

It is a complex process, which mostly happens in the chloroplast of the plant. The chloroplast is the main part for the conversion of carbon dioxide to carbohydrates (sugars).

What is Colorimeter:

The colorimeter works based on Beer-Lambert's law, according to which the absorption of light transmitted through the medium is directly proportional to the medium concentration. In a colorimeter, a beam of light with a specific wavelength is passed through a solution via a series of lenses, which navigate the colored light to the measuring device. This analyzes the color compared to an existing standard. A microprocessor then calculates the absorbance or percent transmittance. If the concentration of the solution is greater, more light will be absorbed, which can be identified by measuring the difference between the amount of light at its origin and that after passing the solution.

In this project, a portable Hach DR-890 Colorimeter is used to measure the Nitrate and Phosphate concentration in wastewater.

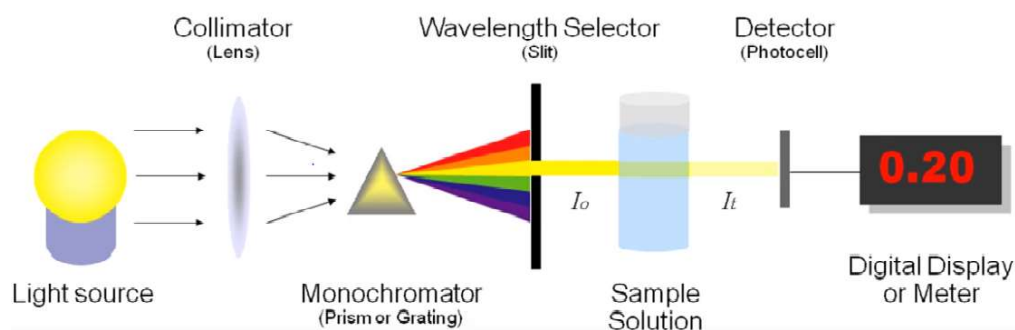


Figure 4: Components of a Colorimeter.

Materials:

1. Micropipette with disposable tips.
2. Clear vinyl tubing with inside diameter 4.8 mm.
3. Fluorescent light.
4. Erlenmeyer flasks 500ml - 22 Nos.
5. Rubber stopper 2-hole - 22 Nos.
6. Glass tube 7.5cm long, 5mm diameter
7. Graduated cylinder 100ml
8. Glass funnel
9. Test tubes.
10. Air pump
11. Five way control valve – 2 Nos.
12. Two way air separators – 1 No.
13. Air stones - 3 packs.
14. 4 specimen algae culture solution:
 - i. *Chlamydomonas reinhardtii*
 - ii. *Arthrospira platensis* (Spirulina)
 - iii. *Scenedesmus quadricauda*
 - iv. *Chlorella vulgaris*.
15. Wastewater sample – 6 L.
16. Distilled Water – 1 L.
17. Whatman No. 1 filter paper.
18. Portable Hach DR-890 colorimeter.
19. Hach Nitra Ver 5 Nitrate reagent powder pillow.
20. Phosphate reagent set for Phosphate Low Range tests.
21. Compound microscope.
22. HDCE-X5 microscope digital camera
23. Digital camera for recording the steps.
24. Laptop/computer for Data recording and analysis.

Procedures:

1. Collection of Wastewater

- a) For purpose of this project, Instead of using the actual Wastewater from treatment plants due to rules and regulations on handling wastewater, we simulated the wastewater by mixing fertilizer Ammonium Nitrate and Diammonium Phosphate with distilled water.
- b) Using Whatman No. 1 filter paper, filter the wastewater to remove suspended solid particles.

2. Collection of different Algae specimen:

- a) For this project, collected different specimens of algae specimen that are i) Readily available ii) Easy to grow/culture. iii) Does not have adverse effect on environment and human being. iv) Is known to be pollution tolerant and having absorption capacity and thus effective in remediation of wastewater.
- b) There are many different specimen of algae available, like Chlorella, Ankistrodesmus, Scenedesmus, Euglena, Chlamydomonas, Arthrospira, Oscillatoria, Micractinium and Golenkinia.
- c) Based on the abovementioned criteria, the following specimen of algae are used in the project:
 - i. Chlamydomonas reinhardtii
 - ii. Arthrospira platensis (Spirulina)
 - iii. Scenedesmus quadricauda
 - iv. Chlorella vulgaris.

3. Experiment Setup

a) Bioreactor for the algae bioremediation:

To allow a healthy environment for the algae to grow and bioremediate nitrates and phosphates, I made bioreactors using Erlenmeyer flasks and an air pump to provide a source of carbon dioxide in the flasks.

- a. Take a 2-holed rubber stopper and fit a small glass tubes into each hole.
- b. Cut a piece of the plastic tube as long as the flask. Connect one end of this tube to the small glass tube piece.
- c. In the other end of this tube, fit an air stone. Replace the air stone every 2-3 days because the algae can clog up the stone pores and stop air flow.
- d. In one of the holes, put a small piece of a cotton ball to cover up the hole. This is to prevent contamination from airborne bacteria into the algae culture.

Repeat this setup for each types of algae selected.
- e. Cut another piece of the plastic tube as long as needed so the flask stopper can be connected to the control valve. Repeat this for each flask.
- f. Connect the air control valve with 2-way air separator (or T-valve).
- g. Cut another piece of the plastic tube long enough to connect the air pump to the 2-way separator.
- h. Turn the valves in open position to check if the pump is working.

b) Control Solution:

The filtered and untreated wastewater is used as control solution. 200 ml of filtered wastewater is collected in 500 ml sterilized Erlenmeyer flask.

c) 4 Algae Specimen solutions:

- a. Take 200 ml of filtered wastewater in a bioreactor flask described in step 3 a.
- b. Add 4 ml of the first algae stream into the wastewater.
- c. Cover the flask with the rubber stopper prepared before and connect each flask to the set-up for the air intake and start the pump.
- d. Keep a fluorescent light on for a fast photosynthesis of green algae.
- e. Repeat this for each of the 4 algae specimens.

d) Number of Trials:

For this project, I repeated the test in 5 different trials using the same control and specimen solutions.

e) Controlled environment:

The test is conducted in a under controlled conditions (temperature $72 \pm 2^{\circ}\text{F}$) for a total duration of 10 days.

4. Measuring Nitrate and Phosphate Concentration:

Nitrate Concentration:

- a) Using a portable Hach DR-890 colorimeter, measure the nitrate concentration of the solution by cadmium reduction method (Hach Method 8039). In this method, the cadmium metal reduces nitrates present in the sample to nitrite. This nitrite form an amber-colored product by reacting in acidic medium with Sulfanilic acid. This amber color indicates the presence of nitrate and intensity is converted to (N-NO₃) mg/L or ppm.
- b) Prepare a blank by adding Hach Nitra Ver 5 Nitrate reagent powder pillow to 10 ml of distilled water in a test tube. Use this blank to calibrate the colorimeter to zero prior to

testing of samples.

- c) Using a pipette, collect 2 ml of the solution (Control or the specimen solution) in a test tube.
- d) Add the Hach Nitra Ver 5 Nitrate reagent powder pillow to the appropriate dilution of the supernatant. Final volume of the testing sample should be 10ml. Shake the sample for one minute and allow to rest for five minutes permitting the reaction to complete.
- e) Insert the sample test tube to the colorimeter and take the concentration measurement.

Phosphate Concentration:

- a) Similar to the Nitrate measurement, using a portable Hach DR-890 colorimeter, measure the phosphate concentration of the solution by the molybdate-ascorbic acid method (Hach Method 8048). Orthophosphate present in the sample reacts with molybdate in the reagent in the acid medium to produce a phospho-molybdate complex. Ascorbic acid reduces this complex and forms a blue color. This blue color indicates the presence of phosphate and the intensity is converted to (P-PO₄) mg/L or ppm.
- b) Prepare a blank by adding Phosphate reagent powder pillow to 10 ml of distilled water in a test tube. Use this blank to calibrate the colorimeter to zero prior to testing of samples.
- c) Using a pipette, collect 2 ml of the solution (Control or the specimen solution) in a test tube.
- d) Add the Phosphate reagent powder pillow to the appropriate dilution of the supernatant. Final volume of the testing sample should be 10ml. Shake the sample for 15 seconds and allow to rest for 2 minutes permitting the reaction to complete.
- e) Insert the sample test tube to the colorimeter and take the concentration measurement.

5. Test procedure

- a.** At first, measure the Nitrate and Phosphate concentration of the Control solution following the process described in Step 4 (a and b). This gives the amount of Nitrate and Phosphate present in the filtered wastewater.
- b.** Using the experiment setup as described in Step 3c, setup the 4 specimen algae solution in the bioreactor. Keep the bioreactor for the entire duration of the project (10 Days).
- c.** Every day take a 2 ml sample from the bioreactor Erlenmeyer Flasks of each specimen algae solution, and measure the Nitrate and Phosphate concentration of the solution following the process described in Step 4 (a and b). Repeat this for all the days of the experiment (10 days).
- d.** Repeat this step for each of the 5 trials.
- e.** Collect the Nitrogen and Phosphate concentration data in excel sheet for analysis.
- f.** Analyze the data collected for results.

Data Table(s) and graph(s):

Nitrate Concentration in Wastewater (mg/L)

Trial -1. Nitrate Concentration (in mg/L)

Days	Control Solution	Sp-1 (C. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	182.88	182.88	182.88	182.88	182.88
1	175.30	130.20	152.32	162.20	168.56
2	167.44	88.42	125.41	128.42	135.42
3	160.45	67.00	81.20	102.45	98.80
4	156.28	42.42	54.12	82.42	72.44
5	152.00	25.01	30.48	55.01	35.51
6	151.30	17.50	22.30	32.74	18.56
7	149.42	13.20	15.60	23.80	16.20
8	146.73	10.80	12.86	16.34	12.86
9	142.65	9.60	11.93	13.20	9.66
10	140.00	7.20	9.40	10.80	8.20

Trial -2. Nitrate Concentration (in mg/L)

Days	Control Solution	Sp-1 (C. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C.vulgaris)
0	180.50	180.50	180.50	180.50	180.50
1	178.23	143.20	160.23	162.20	188.74
2	169.88	95.23	132.25	146.25	165.25
3	161.27	74.62	98.78	110.24	100.02
4	157.39	54.32	65.85	79.36	72.44
5	152.30	25.01	45.25	58.69	35.51
6	151.00	18.90	22.30	42.75	20.58
7	148.73	14.65	18.65	23.80	16.20
8	146.20	11.86	13.56	18.45	14.72
9	142.49	10.63	12.43	14.44	11.28
10	138.05	9.42	10.65	10.98	9.80

Trial -3. Nitrate Concentration (in mg/L)

Days	Control Solution	Sp-1 (C. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	178.25	178.25	178.25	178.25	178.25
1	175.23	152.32	152.32	162.20	162.20
2	169.88	98.65	125.41	128.42	146.25
3	165.05	74.28	81.20	98.78	110.24
4	161.43	42.42	54.12	65.85	79.36
5	158.02	28.65	35.51	45.25	35.51
6	155.45	20.41	20.58	32.74	18.56
7	150.65	13.20	16.20	18.65	16.20
8	144.09	11.23	12.86	13.56	12.86
9	140.58	9.82	11.65	11.20	11.20
10	136.52	6.80	9.40	10.65	9.40

Trial -4. Nitrate Concentration (in mg/L)

Days	Control Solution	Sp-1 (C. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	188.50	188.50	188.50	188.50	188.50
1	180.88	130.20	152.32	162.20	168.56
2	172.56	98.65	125.41	128.42	135.42
3	166.64	65.85	81.20	102.45	82.42
4	159.81	42.42	54.12	82.42	49.85
5	154.61	25.01	30.48	55.01	35.51
6	150.09	17.50	22.30	32.74	22.30
7	149.38	13.20	15.60	23.80	18.65
8	146.45	10.80	13.20	16.34	10.80
9	143.73	9.85	11.82	13.88	9.66
10	141.79	8.50	9.65	11.42	9.05

Trial -5. Nitrate Concentration (in mg/L)

Days	Control Solution	Sp-1 (c. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	186.28	186.28	186.28	186.28	186.28
1	179.30	162.20	152.32	156.00	154.36
2	171.25	128.42	128.42	119.98	125.43
3	168.25	102.45	102.45	105.24	100.58
4	162.51	54.12	82.42	88.78	88.73
5	158.30	30.48	35.51	55.01	42.91
6	154.79	22.30	22.30	32.74	24.57
7	149.88	13.20	18.65	23.80	16.20
8	147.19	11.28	13.88	16.34	12.86
9	145.07	10.25	12.57	14.82	10.66
10	143.90	7.86	10.68	11.45	9.53

Average of 5 trials:

Days	Control Solution	Sp-1 (c. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	183.28	183.28	183.28	183.28	183.28
1	177.79	143.62	153.90	160.96	166.48
2	170.20	101.87	127.38	130.30	141.55
3	164.33	76.84	88.97	103.83	98.41
4	159.48	47.14	62.13	79.77	72.56
5	155.05	26.83	35.45	53.79	36.99
6	152.53	19.32	21.96	34.74	20.91
7	149.61	13.49	16.94	22.77	16.69
8	146.13	11.19	13.27	16.21	12.82
9	142.90	10.03	11.93	13.51	10.49
10	140.05	7.96	9.96	11.06	9.20

Nitrate Remediation Efficiency:

$$Ei = \{(Co - Ci) / Co\} * 100$$

where Co and Ci were the concentrations of contaminant at the start of experiment (day 0) and day i , respectively.

Days	Control Solution	Sp-1 (c. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	0.00	0.00	0.00	0.00	0.00
1	3.00	21.64	16.03	12.18	9.16
2	7.14	44.42	30.50	28.91	22.77
3	10.34	58.08	51.46	43.35	46.31
4	12.98	74.28	66.10	56.48	60.41
5	15.40	85.36	80.66	70.65	79.82
6	16.78	89.46	88.02	81.04	88.59
7	18.37	92.64	90.76	87.58	90.89
8	20.27	93.89	92.76	91.16	93.01
9	22.03	94.53	93.49	92.63	94.28
10	23.59	95.66	94.57	93.97	94.98

Phosphate Concentration in Wastewater (mg/L)

Trial -1. Phosphate Concentration (in mg/L)

Days	Control Solution	Sp-1 (c. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	26.21	26.21	26.21	26.21	26.21
1	25.40	20.32	22.90	21.32	21.77
2	23.88	15.74	18.78	17.34	16.45
3	22.50	12.98	14.38	14.75	13.40
4	22.11	10.30	12.30	11.90	11.75
5	21.80	8.55	9.55	7.55	10.85
6	21.04	6.88	7.88	6.80	8.88
7	20.18	6.21	6.98	6.05	7.25
8	19.72	5.30	6.58	4.60	5.92
9	18.95	4.68	6.36	2.68	4.68
10	18.15	3.90	6.20	1.88	4.38

Trial -2. Phosphate Concentration (in mg/L)

Days	Control Solution	Sp-1 (c. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	23.20	23.20	23.20	23.20	23.20
1	22.40	18.32	22.90	21.32	21.90
2	21.78	15.20	20.78	20.34	19.45
3	21.50	10.98	19.38	18.75	17.40
4	20.15	9.80	16.30	13.88	11.75
5	19.80	9.21	13.55	10.50	10.85
6	19.04	8.88	12.88	8.60	9.89
7	18.18	7.75	10.90	7.05	8.25
8	17.72	6.50	9.45	5.50	6.45
9	16.75	6.25	9.04	3.68	5.68
10	15.15	5.90	8.20	2.56	4.30

Trial -3. Phosphate Concentration (in mg/L)

Days	Control Solution	Sp-1 (c. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	25.20	25.20	25.20	25.20	25.20
1	24.40	22.40	23.50	22.90	18.32
2	23.88	21.78	21.88	20.78	15.20
3	22.50	21.50	21.35	19.38	10.98
4	22.11	10.30	20.34	16.30	9.80
5	21.80	9.25	18.75	12.98	10.85
6	20.15	8.67	13.88	10.30	8.88
7	19.80	6.68	9.05	8.55	7.25
8	19.04	6.20	7.90	6.88	5.92
9	18.18	4.68	6.36	6.21	4.68
10	16.15	3.90	5.96	3.87	4.38

Trial -4. Phosphate Concentration (in mg/L)

Days	Control Solution	Sp-1 (c. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	28.75	28.75	28.75	28.75	28.75
1	28.40	24.33	22.90	24.33	21.77
2	27.88	23.20	18.78	23.20	16.45
3	27.25	21.32	14.38	22.90	13.40
4	26.61	17.34	12.30	20.78	11.75
5	25.40	14.75	9.55	19.38	10.55
6	23.88	11.90	7.88	16.30	8.92
7	22.50	10.98	6.98	10.85	6.55
8	22.11	9.80	6.58	8.88	5.92
9	21.80	9.05	6.36	7.25	4.97
10	21.20	7.50	6.20	5.20	4.74

Trial -5. Phosphate Concentration (in mg/L)

Days	Control Solution	Sp-1 (c. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	20.50	20.50	20.50	20.50	20.50
1	19.80	17.38	19.38	18.75	19.96
2	19.04	16.30	18.78	13.88	19.02
3	18.18	14.75	14.38	10.50	18.32
4	18.15	12.65	12.30	8.60	15.20
5	17.80	10.98	10.98	7.75	10.55
6	17.04	8.75	9.80	7.32	8.92
7	16.75	7.75	8.40	7.00	6.55
8	16.05	6.50	8.04	6.50	5.50
9	15.45	6.25	7.05	6.25	4.75
10	15.03	5.90	6.44	4.80	3.92

Average of All trials:

Days	Control Solution	Sp-1 (c. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	24.772	24.77	24.77	24.77	24.77
1	24.08	20.55	22.32	21.72	20.74
2	23.292	18.44	19.80	19.11	17.31
3	22.386	16.31	16.77	17.26	14.70
4	21.826	12.08	14.71	14.29	12.05
5	21.32	10.55	12.48	11.63	10.73
6	20.23	9.02	10.46	9.86	9.10
7	19.482	7.87	8.46	7.90	7.17
8	18.928	6.86	7.71	6.47	5.94
9	18.226	6.18	7.03	5.21	4.95
10	17.136	5.42	6.60	3.66	4.34

Phosphate Remediation Efficiency:

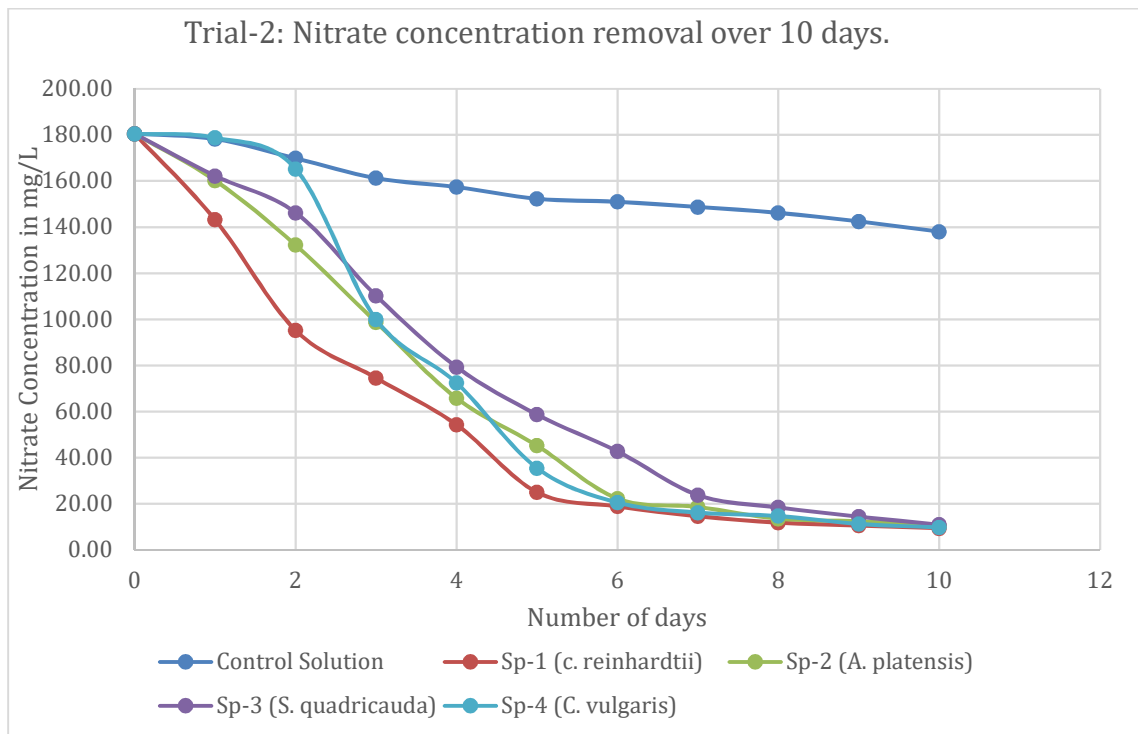
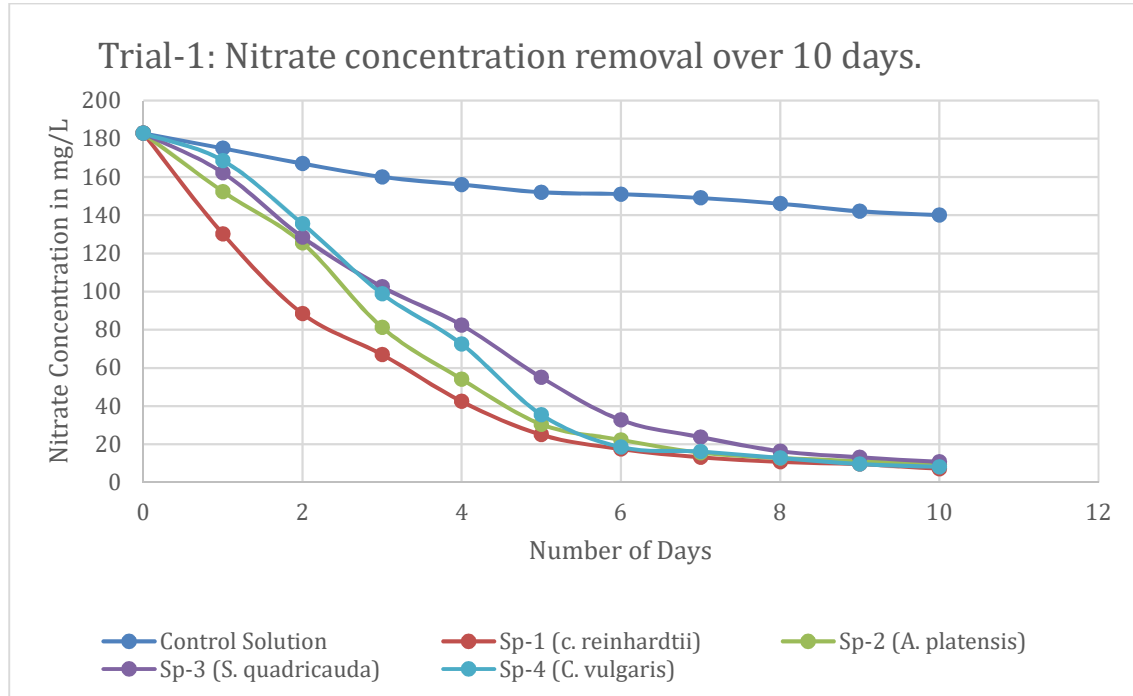
$$Pi = \{(Qo - Qi) / Qo\} * 100$$

where Qo and Qi were the concentrations of contaminant at the start of experiment (day 0) and day i , respectively.

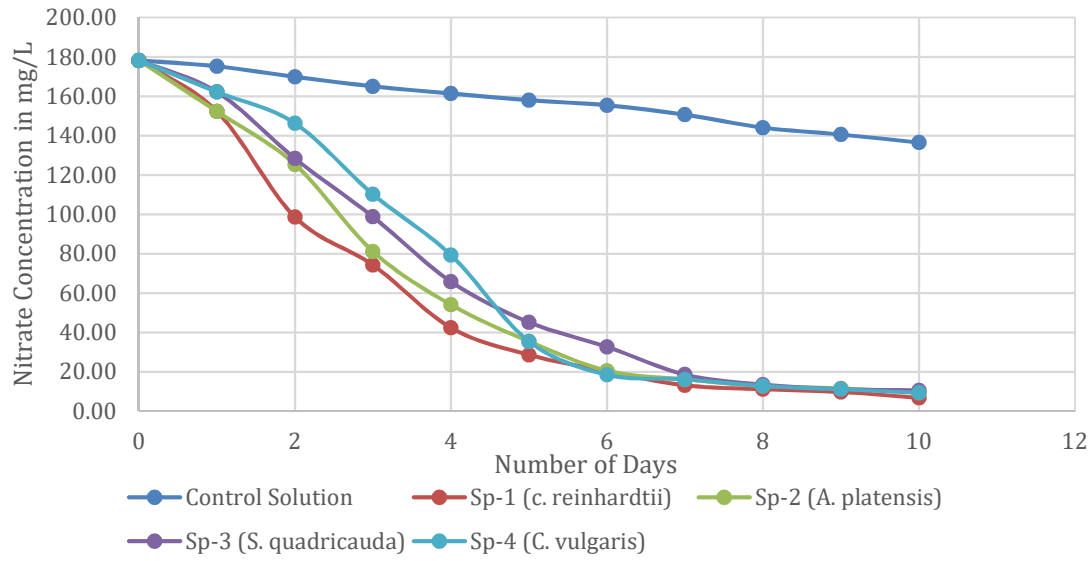
Days	Control Solution	Sp-1 (c. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	0.00	0.00	0.00	0.00	0.00
1	2.79	17.04	9.91	12.30	16.26
2	5.97	25.54	20.07	22.86	30.11
3	9.63	34.18	32.29	30.34	40.66
4	11.89	51.24	40.63	42.31	51.36
5	13.94	57.42	49.64	53.04	56.68
6	18.34	63.60	57.76	60.18	63.27
7	21.35	68.21	65.84	68.11	71.06
8	23.59	72.31	68.88	73.87	76.01
9	26.42	75.04	71.61	78.95	80.01
10	30.83	78.12	73.36	85.22	82.46

Graph(s):

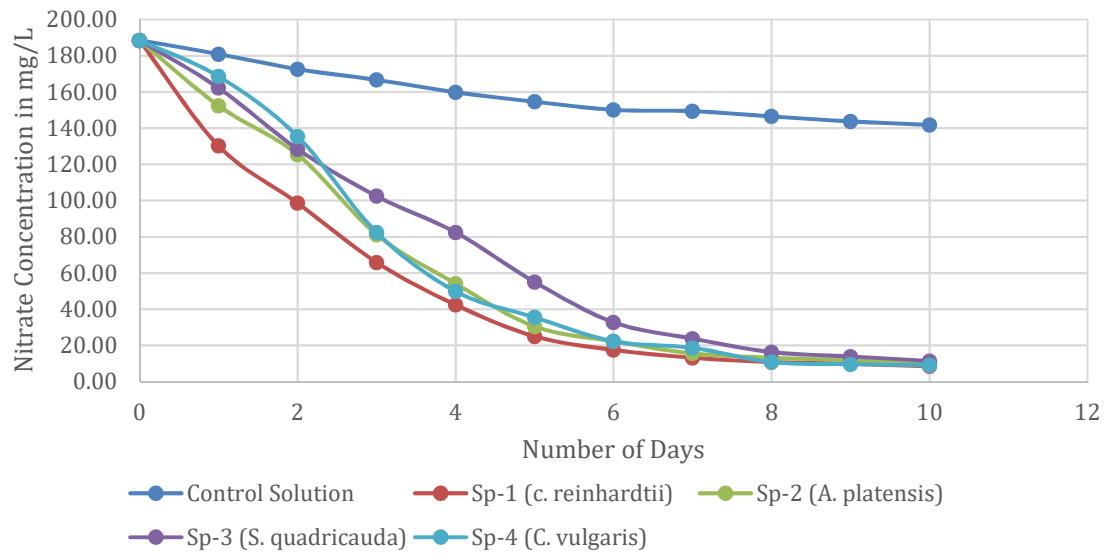
Nitrate Concentration in Wastewater (In mg/L)



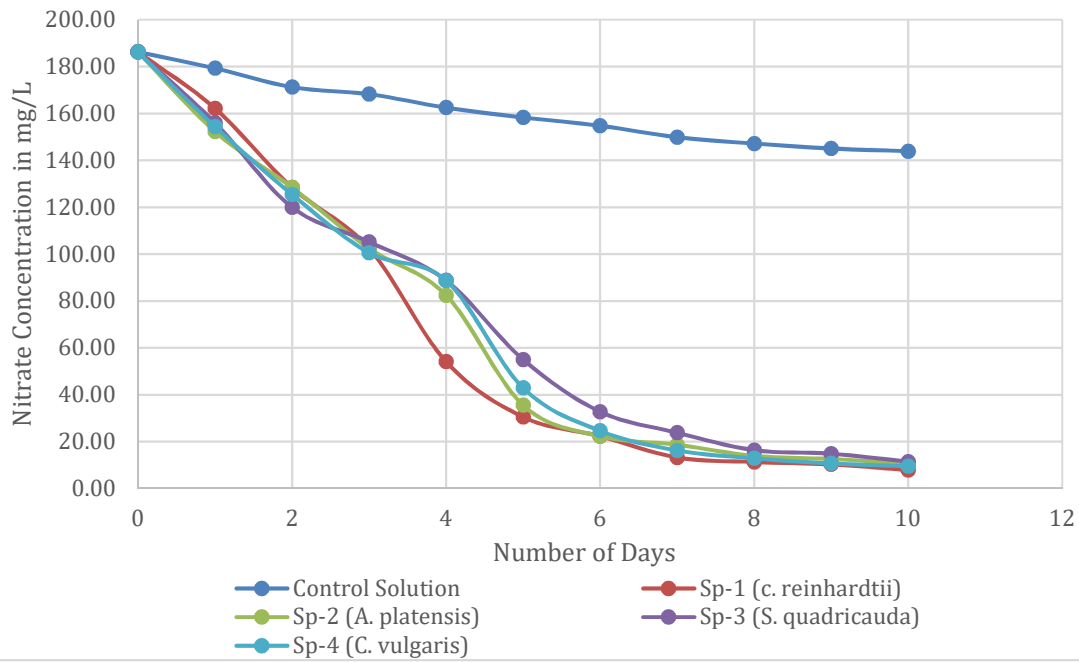
Trial-3: Nitrate concentration removal over 10 days.



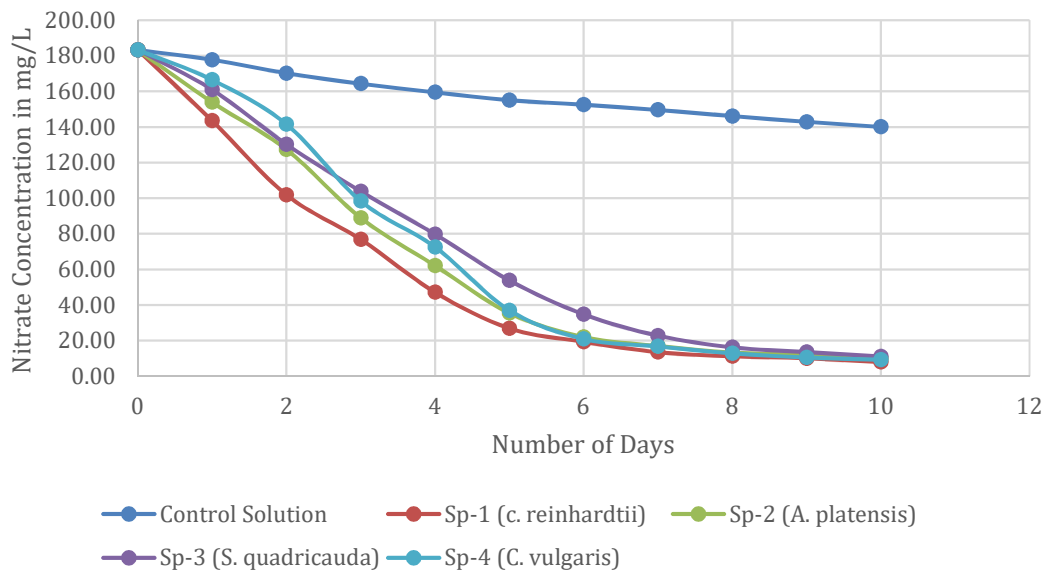
Trial-4: Nitrate concentration removal over 10 days.



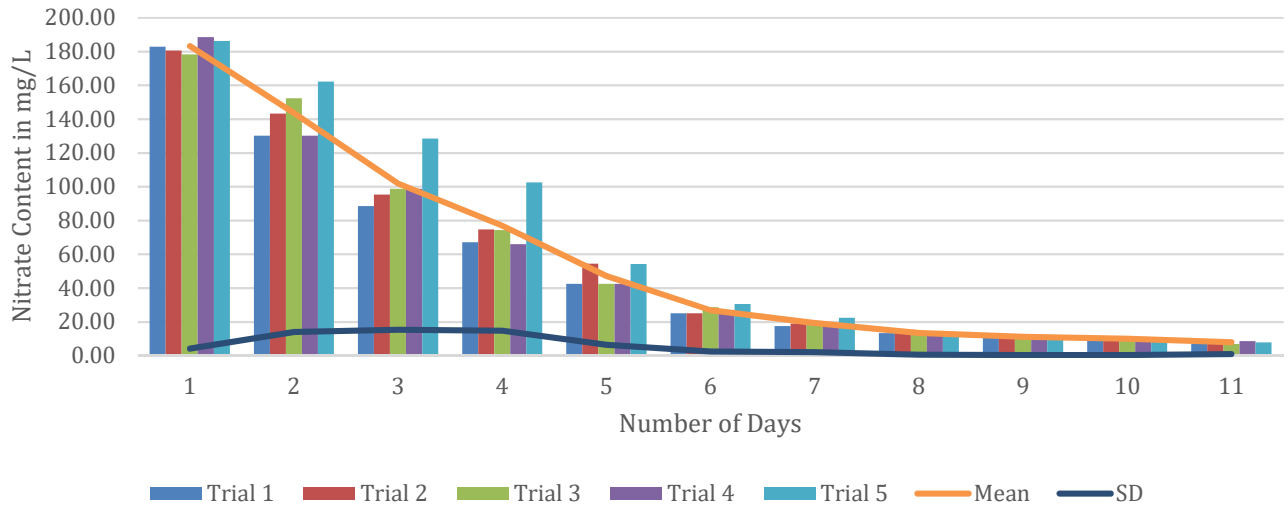
Trial-5: Nitrate concentration removal over 10 days.



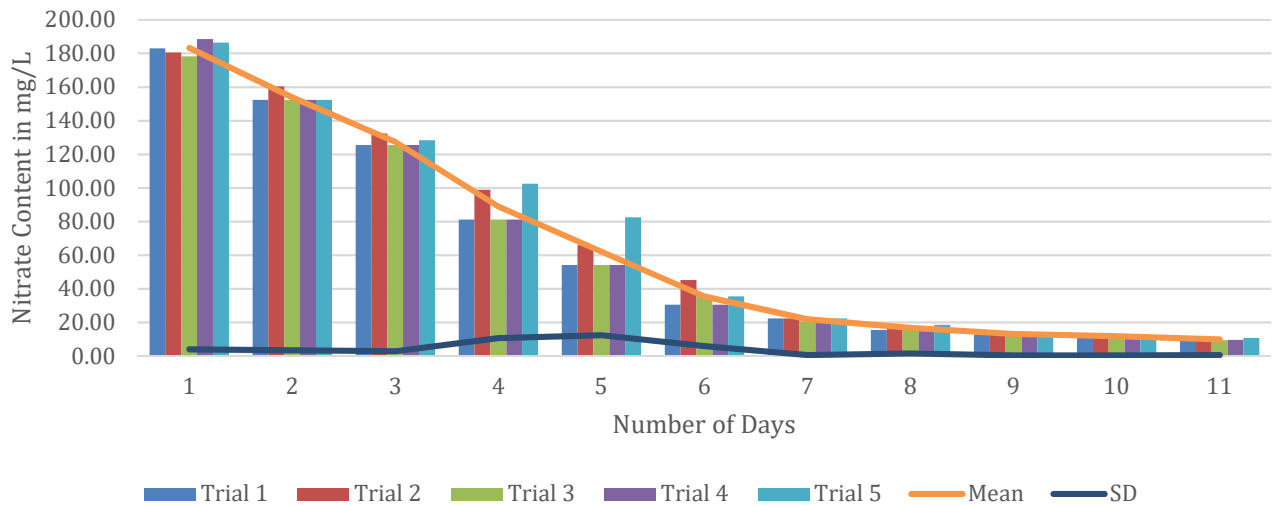
Average of All Trials:

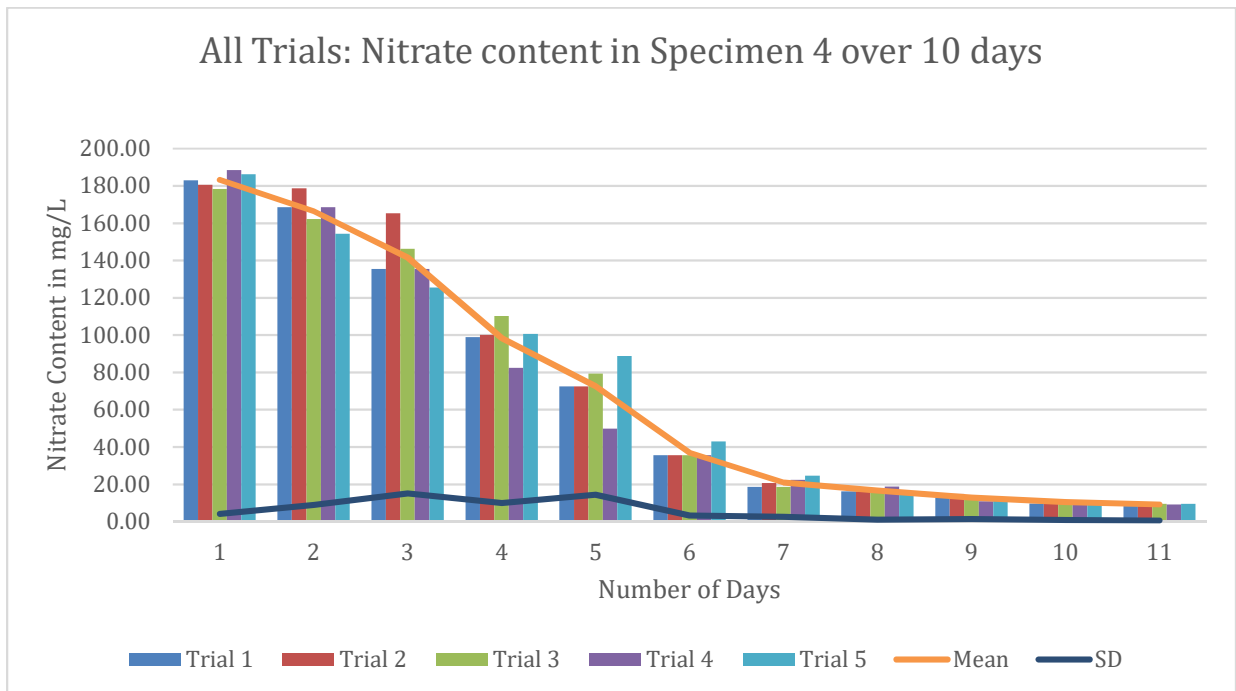
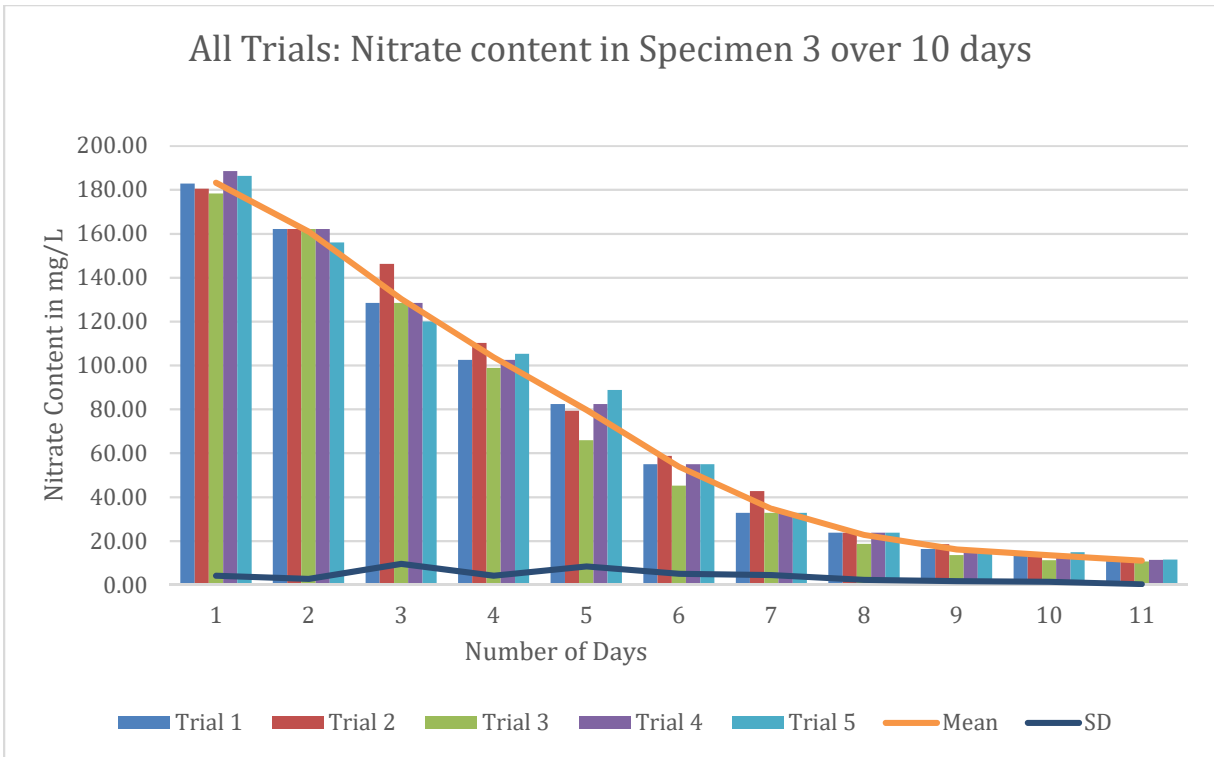


All Trials: Nitrate content in Specimen 1 over 10 days

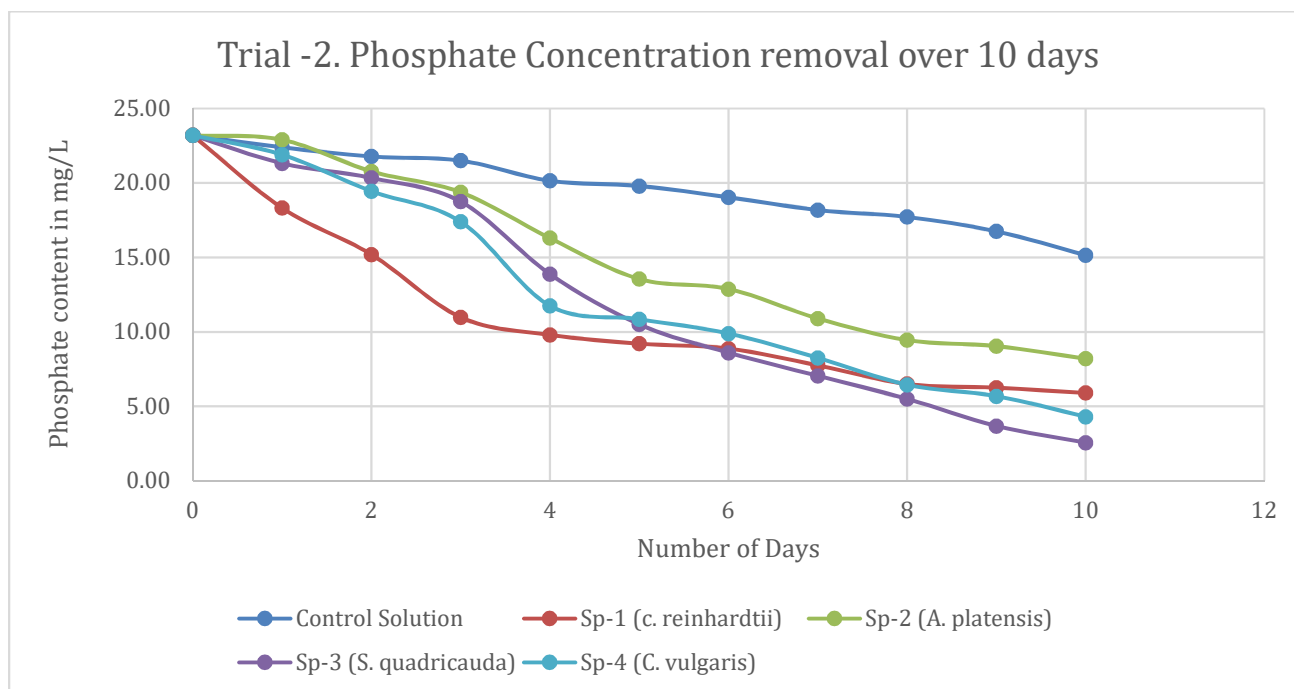
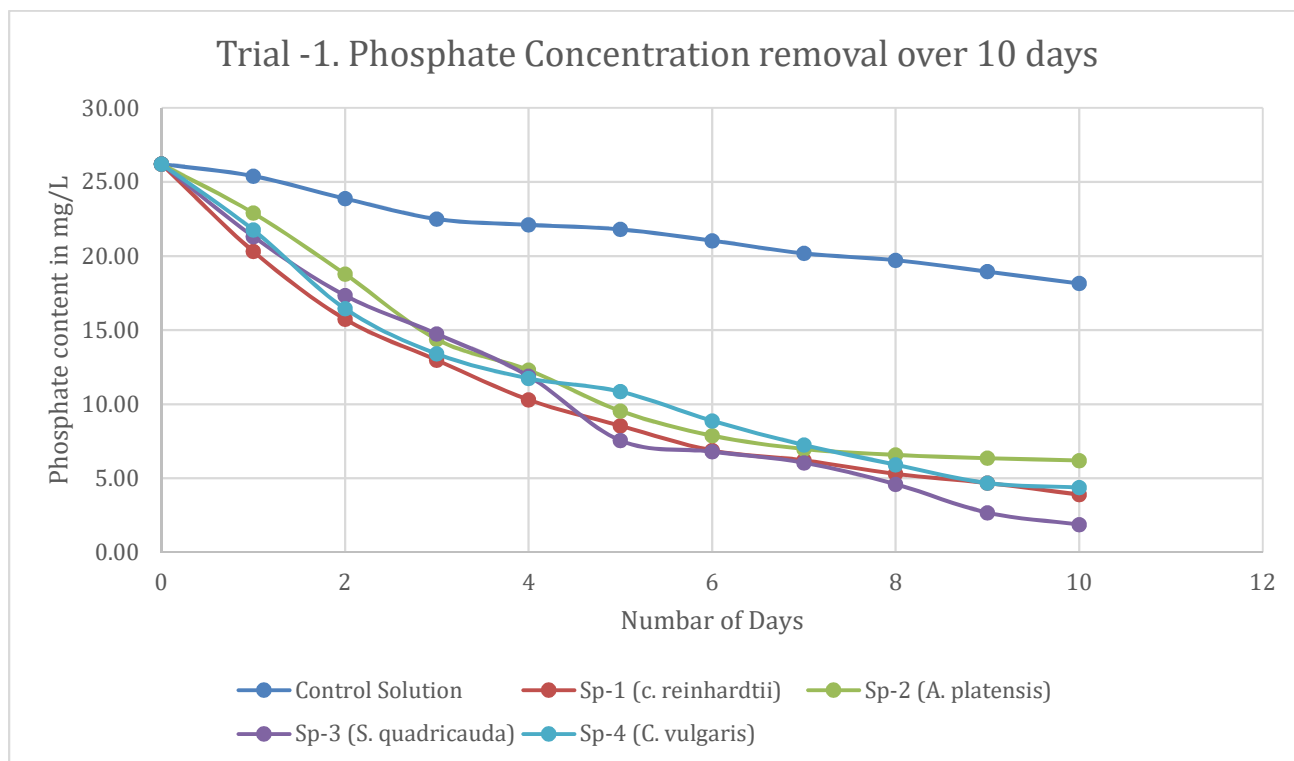


All Trials: Nitrate content in Specimen 2 over 10 days

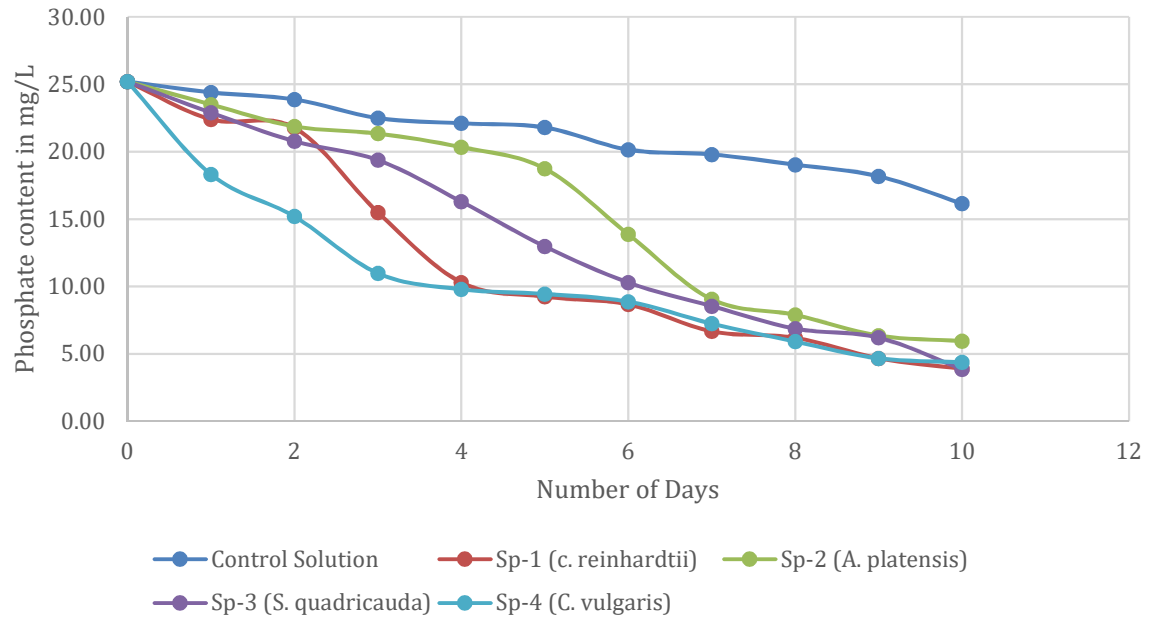




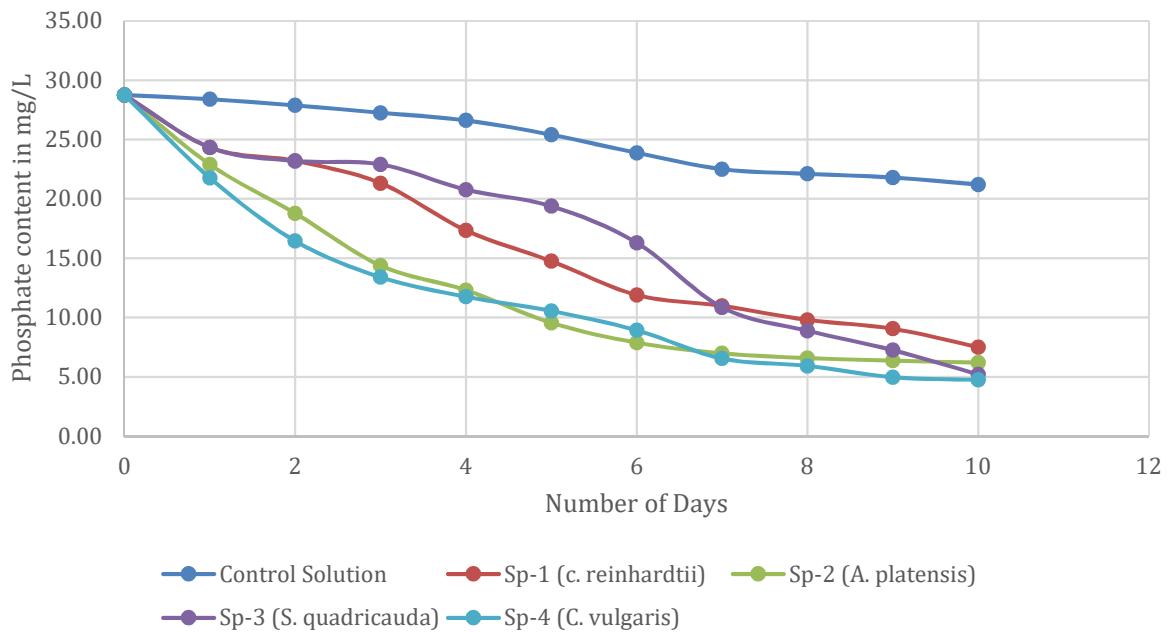
Phosphate Concentration in Wastewater (In mg/L)



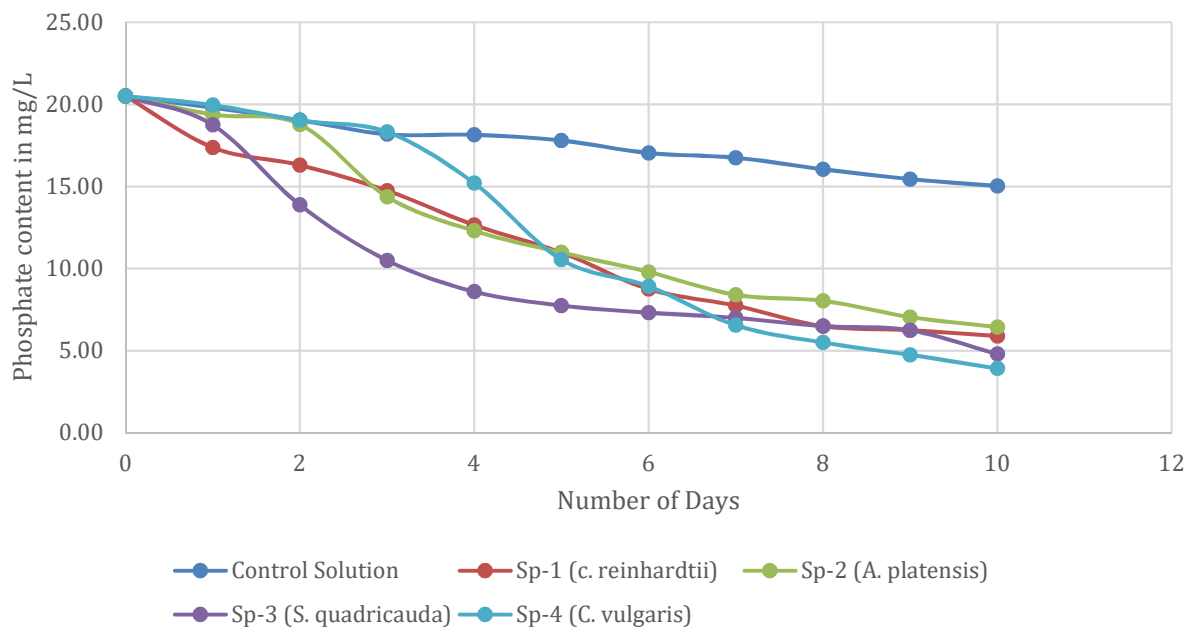
Trial -3. Phosphate Concentration removal over 10 days



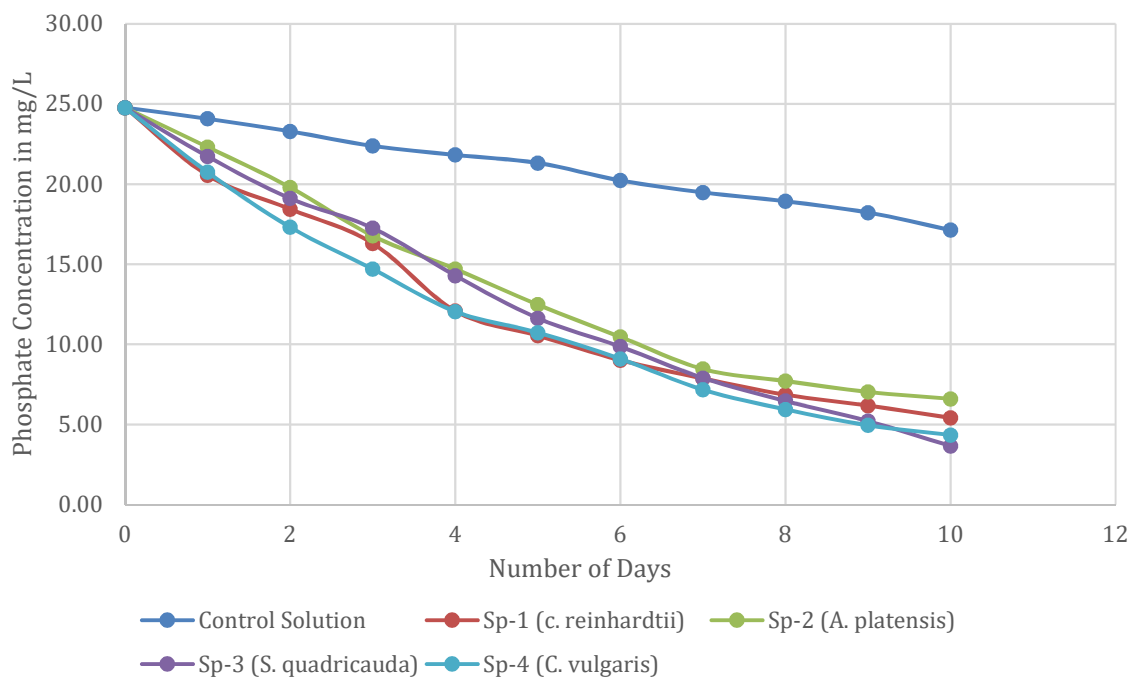
Trial -4. Phosphate Concentration removal over 10 days

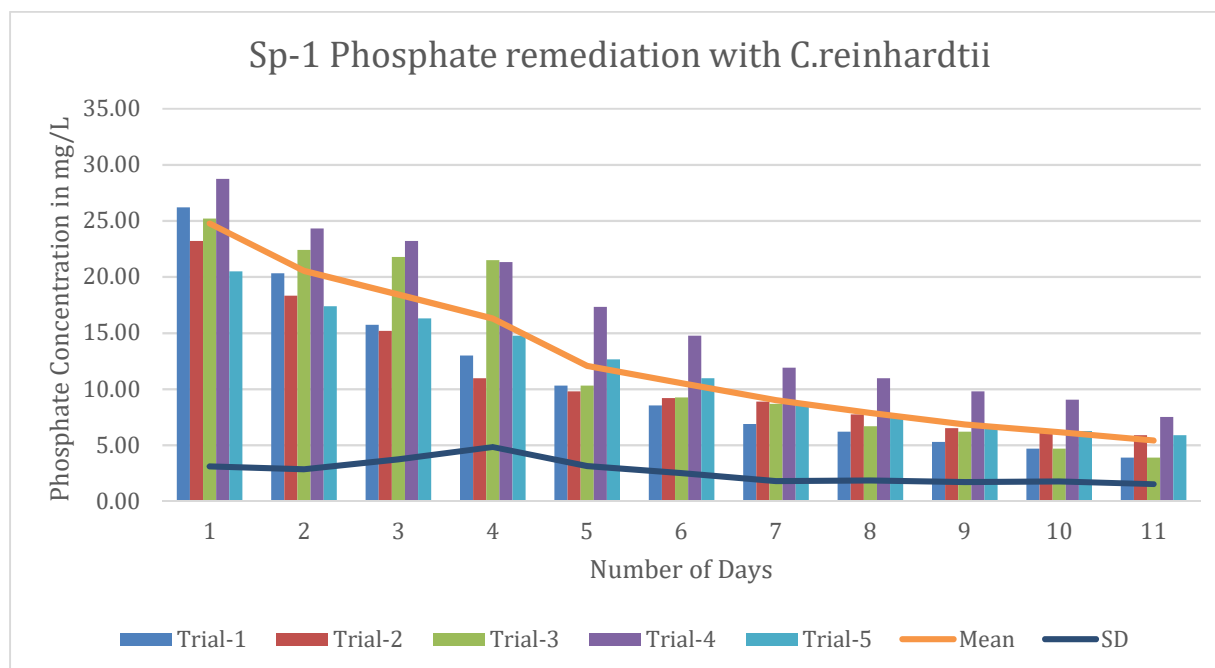
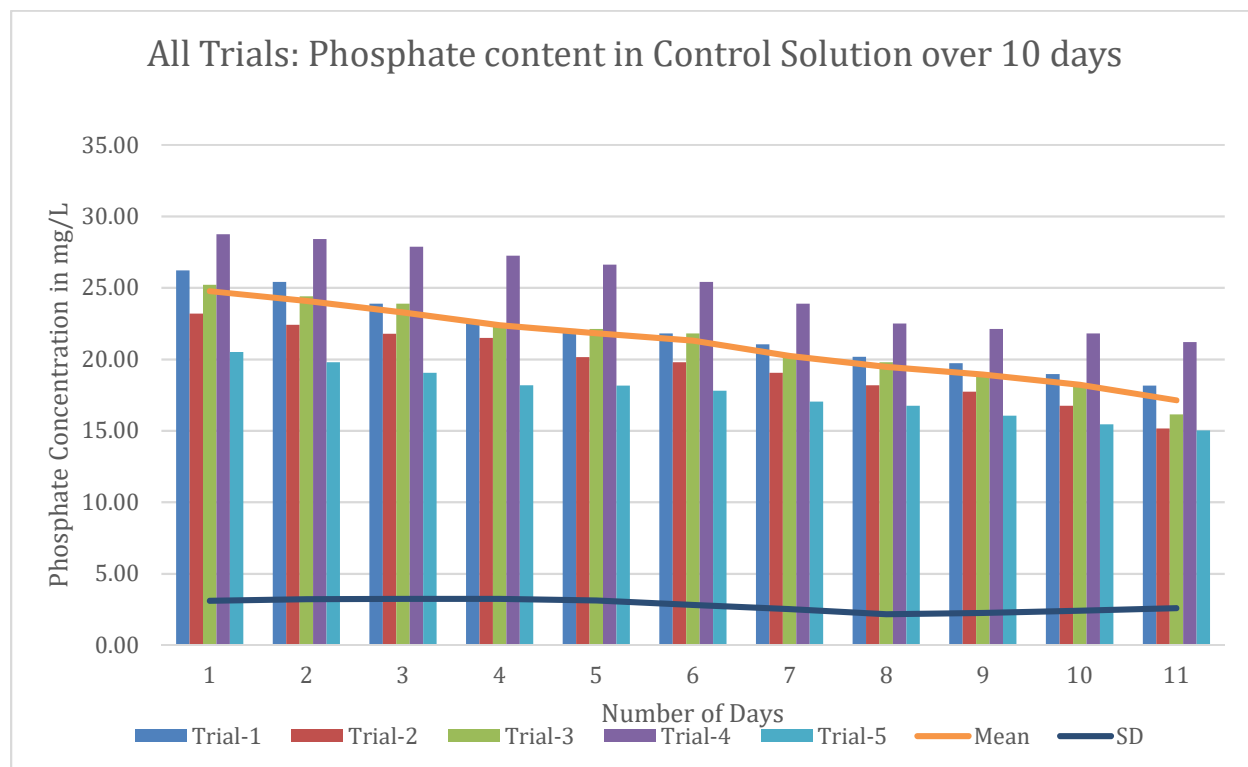


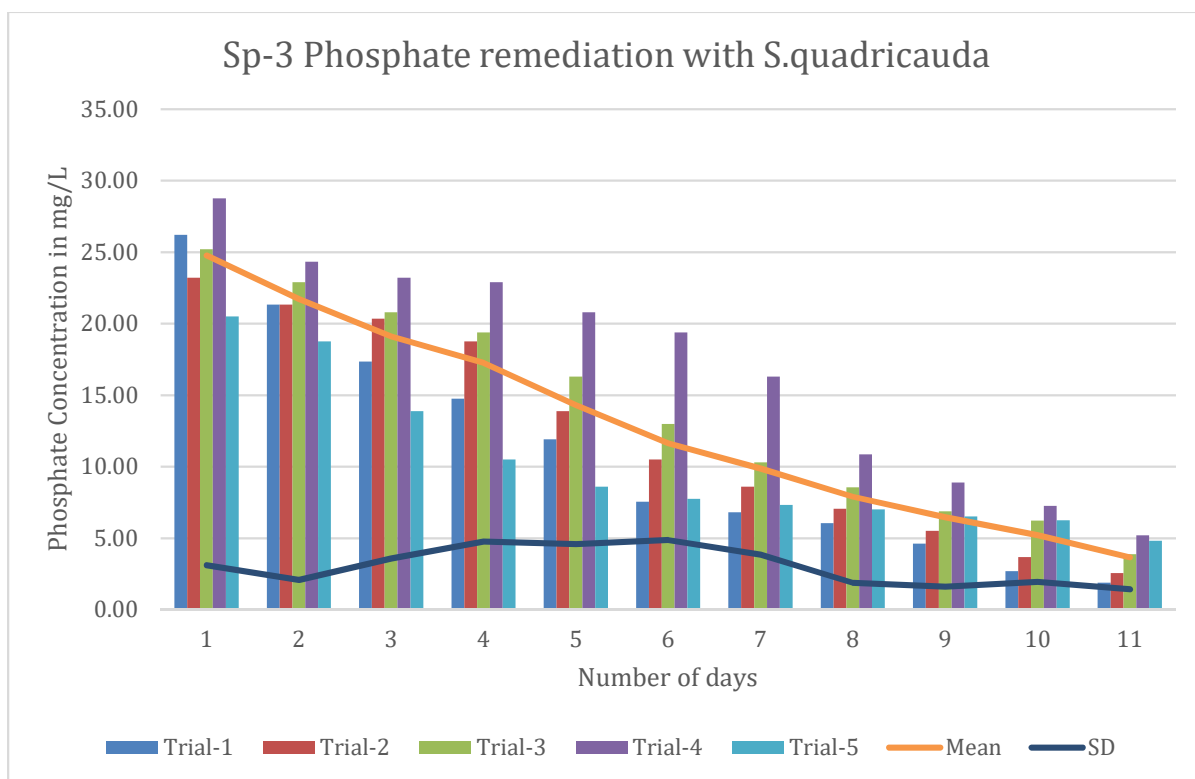
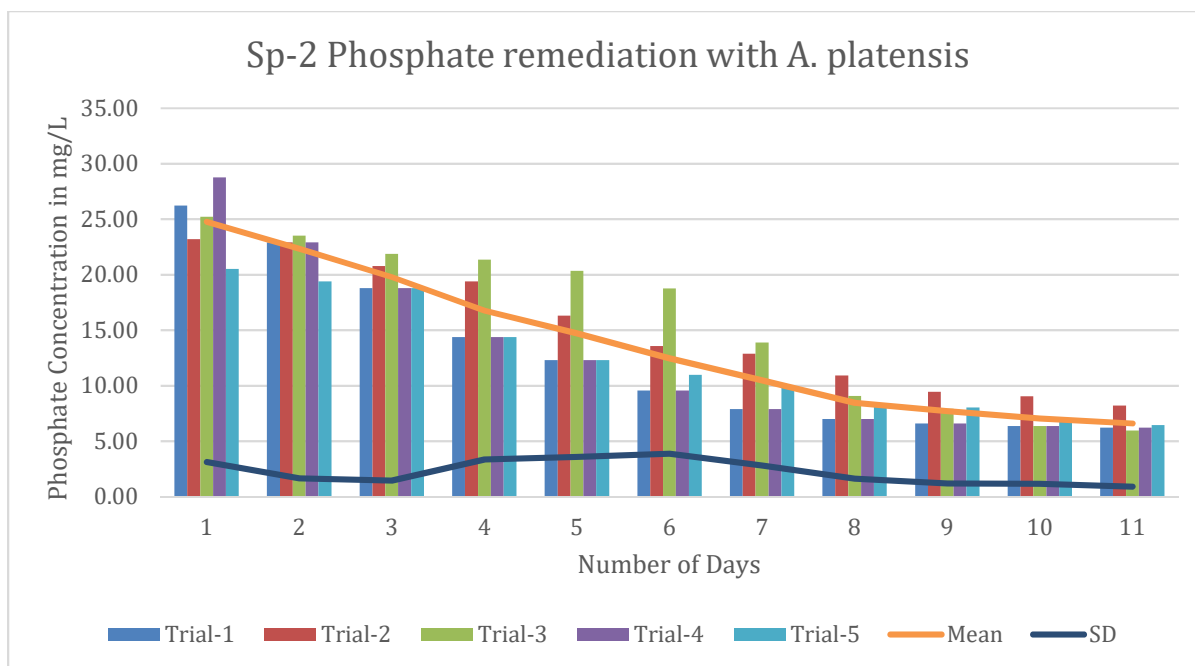
Trial -5. Phosphate Concentration removal over 10 days

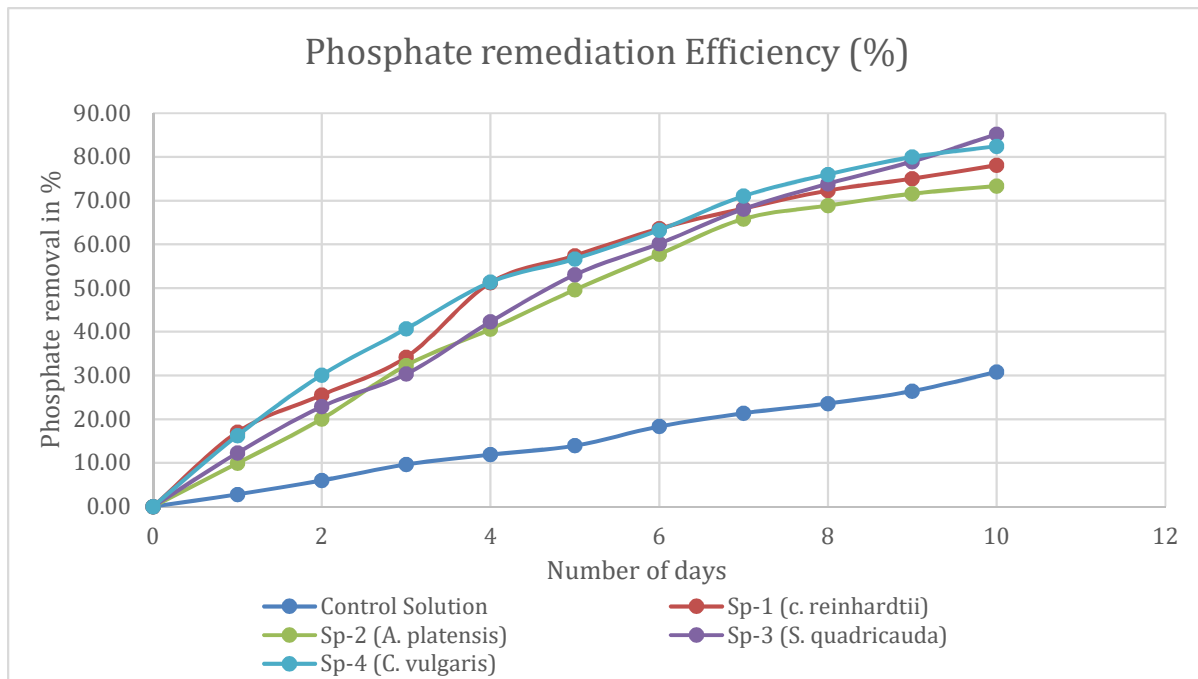
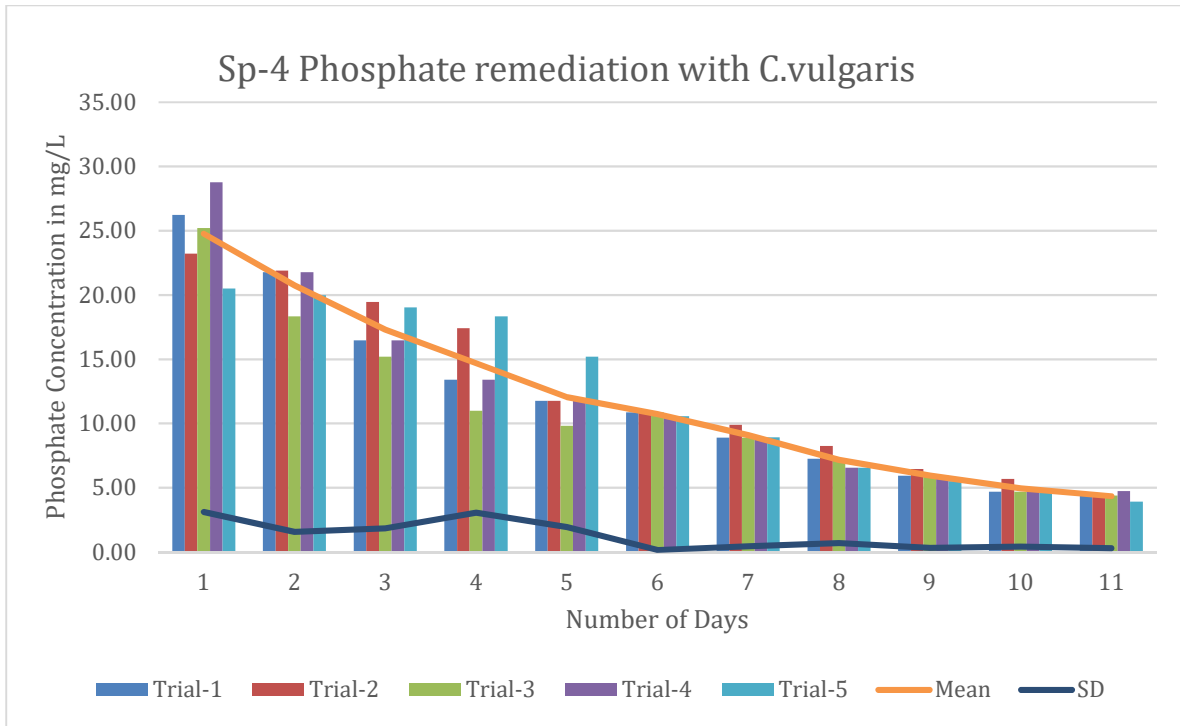


Average of all trials









Data Analysis:

The original purpose of this experiment was to find out how effective different strains of algae are in bioremediation of Nitrate and Phosphate from wastewater. Here, the Control solution used was the filtered wastewater. The 4 specimen solutions were wastewater mixed with different specimens of algae. The four specimen are:

- i. Specimen 1: *Chlamydomonas reinhardtii*
- ii. Specimen 2: *Arthrospira platensis* (Spirulina)
- iii. Specimen 3: *Scenedesmus quadricauda*
- iv. Specimen 4: *Chlorella vulgaris*.

Nitrate Remediation: To measure Nitrate removal from wastewater by different specimens of algae, I used portable Hach DR-890 colorimeter, measuring the nitrate concentration of the Control solution and four different specimen solutions by cadmium reduction method (Hach Method 8039) every day for a 10 days period. Based on the data collected it is observed that all the 4 types of algae specimens were able to reduce Nitrate concentration from wastewater. The initial concentration of nitrate in the wastewater sample ranged from 178.25 to 188.50 mg/L (Mean: 183.28 mg/L). The 4 algal specimens were able to reduce it to the following: *Chlamydomonas reinhardtii* (7.96 mg/L), *Arthrospira platensis* (9.96 mg/L), *Scenedesmus quadricauda* (11.06 mg/L) and *Chlorella vulgaris* (9.20 mg/L). In essence, the average percentage nitrate removal efficiency achieved by the four specimens were *Chlamydomonas reinhardtii* (95.66%), *Arthrospira platensis* (94.57%), *Scenedesmus quadricauda* (93.97%) and *Chlorella vulgaris* (94.98%).

The Nitrate remediation/uptake efficiency of the algal specimen was calculated by the formula below:

$$Ei = \{(Co - Ci) / Co\} * 100$$

where C_0 and C_i were the concentrations of contaminant at the start of experiment (day 0) and day i , respectively.

Hence it can be concluded that the *Chlamydomonas reinhardtii* specimen is able to achieve the most nitrate removal with 95.66%, and all the specimens were able to remediate nitrate to the range of 93-96%. The EPA prescribed limit for nitrate concentration in water is maximum 10 mg/L. 3 of the 4 algal specimens were able to bring the nitrate concentration to that level over a period of 10 days.

Phosphate Remediation: To measure the Phosphate removal from wastewater by different specimens of algae, I used portable Hach DR-890 colorimeter, measuring the phosphate concentration of the Control solution and four different algal specimen solutions by the molybdate-ascorbic acid method (Hach Method 8048).

Based on the resulted data, all 4 of the algal species efficiently removed phosphorus from the wastewater. The rate and degree of phosphorus uptake from the wastewater by each of the six species is presented in Figure---. In all 4 of the bioreactors, prior to treatment with algae, the initial phosphorus content was on average 24.77 mg/L. Over the period of 10 days of experiment, the phosphorus concentration was reduced to maximum 3.66 mg/L. Phosphate remediation efficiency by *Scenedesmus quadricauda* was more than other species during the experimental period. 85.22% of initial phosphate content was consumed by *Scenedesmus quadricauda*. It was then followed by *Chlorella vulgaris* (82.46%) *Chlamydomonas reinhardtii* (78.12%) and *Arthrospira platensis* (73.36%). Regardless of the algal specie, by the end of the 10 days period, A range of 73-85 % of the initial wastewater phosphate content was removed by the algal species in each of the reactors.

The Phosphate remediation/ uptake efficiency of the algal specimen was calculated by the formula below:

$$P_i = \{(Q_0 - Q_i) / Q_0\} * 100$$

where Q_0 and Q_i were the concentrations of contaminant at the start of experiment (day 0) and day i , respectively.

Conclusion:

My hypothesis had been that if different algae specimens such as *Chlamydomonas reinhardtii*, *Arthrospira platensis*, *Scenedesmus quadricauda* and *Chlorella vulgaris* are used for bioremediation of wastewater by removing the Nitrate and Phosphate concentration from wastewater, all of them would be effective in the remediation of Nitrate and Phosphate, while *Chlamydomonas reinhardtii* would be the most effective specimen as it is having the most absorption capability. Based on my observations and test results mentioned above, my hypothesis is partly accepted. The first part that all the algae specimens would be able to remediate nitrate and phosphate is proven. Also, for the Nitrate removal, *Chlamydomonas reinhardtii* is proven to be most effective remediating upto 96% of the Nitrate concentration and making the water allowable to be discharged to environment again.

However, for the Phosphate removal, while all the 4 specimen were effective, *Scenedesmus quadricauda* is proven to be the most effective algae specimen in phosphate concentration remediation from wastewater remediating upto 85.22% of the Phosphate concentration and making the water allowable to be discharged to environment again.

Therefore, this project concludes that algal bioremediation is a viable alternate technology for treating wastewater by removing the Nitrate and Phosphate concentration in an economical and sustainable way when compared to conventional treatment processes. By this method, the annual wastewater treatment cost to remove Nitrate and Phosphate is much lower and thereby the bioremediation of wastewater using algae provides an effective and environmentally acceptable option for waste water. However, when doing this project, there were many other factors that could have been taken into consideration such as the time taken to react, the controlled temperature and light during the experiment, and the precise measurements of all the samples. It is possible to introduce a source of error in the experiment during sterilization of equipment by contaminating

with previous substances, errors in taking sample measurements etc. Therefore, if I were to do this project again, I would do it on a broader scale, using more controlled environment.

Overall, this was a fun project to execute, and I learnt the importance of wastewater treatment for the environment and how to use sustainable, readily available natural elements such as algae in remediating the harmful elements of wastewater before discharging to the environment.

Future Goal:

In this project I focused on Nitrate and Phosphate removal aspect of bioremediation of wastewater. However, there are many other characteristics of wastewater treatment, that needs to be done, Such as:

- a. BOD and COD remediation.
- b. Removal of toxic Heavy metals such as Cadmium (Cd), Chromium (Cr), Arsenic (As) and Mercury (Hg) etc.
- c. Effect of controlled environment, such as temperature and Light in photosynthesis of Algae.
- d. Introduction of external stimulant such as Magnesium etc. affecting the growth rate of algae and bioremediation of wastewater.
- e. Integrating bioremediation process with the existing wastewater treatment plants assess its performance in more realistic environment.

Therefore, the future goal of my project is to focus on these aspects of wastewater bioremediation.

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